SynapseLex: Neuro-Symbolic Temporal State-Transition Architecture

End-to-End Pipeline: Dynamic Case Graph Stream → Inductive Spectral Attention → Normative Axiomatic Governance → Defeasible Adjudication

STAGE 1: GRAPH FORMATION

§ Authentic Court Judgment

Extracts factual narrative & procedural chronology from Supreme Court gazette (congbobanan.toaan.gov.vn, 2018–2026).

N = 2,840 Judgments | Supreme Court Gazette

⚙ Temporal Event & Entity Parser

Extracts litigants, contractual covenants, breaches, and exact operative dates with strict forward-chaining chronology.

PhoBERT Embed (768d) | Zero Leakage (p=0.00)

☍ Dynamic Case Graph G(t)



Timed Nodes V, Timed Edges E(t)

G(t), Δt

STAGE 2: TEMPORAL ATTENTION

∿ Bochner Spectral Time Kernel

Projects continuous elapsed latency Δt into harmonic Fourier space (R^64). Captures quarantine tolling smoothly.

Dual-Clock: Δt + τ(t) | Fourier Dim: 64

⊞ Axiom-TGAT Multi-Head Attention

Temporal neighborhood inductive routing:

$$a(u,v,t) = Softmax(Q_u (K_v + \Phi(\Delta t)) / \sqrt{d})$$

4 Attention Heads | Inductive Message

▥ Dynamic Case Embedding H(t)

Synthesizes temporal litigant interactions, statutory covenants, and breach delays into continuous contextual embeddings.

Context Tensor: H(t) ∈ R^(N × 128)

H(t)

STAGE 3: NORMATIVE GOVERNANCE

⚖ Normative Hierarchy DAG (NH-DA)

Resolves conflicts by statutory precedence:

- Lex Specialis: Land Law 2024 > BLDS 2015
- Usury Cap: Art. 468 ceiling (20%/year)

Deterministic DAG | Conflict Resolution

⇄ Contract State Automaton (M_state)

4 discrete states under BLDS 2015:

S0: VALID Enforceable (Art 398)	**S1: VOIDABLE** Formal Defect (Art 129)
S2: VOID AB INITIO Prohibited (Art 123)	**S3: EXTINGUISHED** Lapsed >1095d (Art 429)

Non-Monotonic Curing: S1 → S0
if ≥ 2/3 performance completed (Art. 129(2))

⛉ OWL 2 DL Ontology (VCLO-Core)

35 Classes, 233 SROIQ Axioms verified by HermiT Reasoner: Zero Incoherence.

HermiT Checked: 0 Unsatisfiable Classes

s_t, AF

STAGE 4: DEFEASIBLE VERDICT

⚔ Dung Defeasible Head

Resolves asymmetric claim defeat:

A_bar (Procedural)	A_claim (Substantive)
Lapse > 1095d	Deposit Penalty
ACCEPTED	**DEFEATED**

$$GE(AF) = lfp(F_AF) = \{ A_bar \}$$

Grounded Extension guarantees 0 remedy bias

✓ Judicial Verdict (y*)

✓ DISMISS: Statute Lapsed (Art. 429)

✗ Enforce Deposit Penalty (Defeated)

✗ Order Contract Curing (Inapplicable)

Hits@1: 86.41% | Latency: 1.5 ms

∇ Axiomatic Regularization Loss

$$L_total = L_task + \lambda_onto\ L_onto + \lambda_state\ L_state$$

Backpropagates margin violations of HermiT OWL 2 DL axioms into TGAT.

Certified Compliance: 99.12% LCR

Axiomatic Feedback: ∇ L_onto + ∇ L_state (HermiT TBox Consistency)